

BrickAtlas Hierarchy: Crossed Evaluation of Brick Structure, Reasoning, and Executable Intervention

Development Manuscript
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Abstract

Understanding an assembly requires more than recognizing its parts: an agent must reason about dependencies, identify missing information, and execute interventions without violating intermediate constraints. We introduce BrickAtlas Hierarchy, a benchmark that holds object identity fixed while varying these requirements. The Hierarchy-3 release crosses 144 authored digital configurations with 48 shared task families, yielding 6,912 questions on 42,617 visible primitives. Four structural strata each contain 36 configurations; every configuration supports 22 atomic, 15 metacognitive/physical, six procedural, and five integrative tasks. A public state-transition interpreter validates action sequences, while resource and possible-world contracts check schedules and policies. Rapier supplies collision queries and sampled dynamic observations. The same model records and interpreter drive interactive 3D inspection and replay. All 6,912 reference answers satisfy their contracts, and a solver using public preconditions solves all 1,152 action tasks. A historical GPT-4.1 mini pilot achieves 7/16 exact successes; its error decomposition separates interface failures from reasoning errors. These controls establish inspectable execution, while the small pilot, shared constructors, and uncalibrated physics limit performance and real-world claims.

1. Introduction

An agent can identify a gripper yet remove its supporting module too early. It can propose the correct replacement but omit verification, or select a useful inspection without specifying what to do after each possible observation. Such failures couple structural understanding with sequential and conditional reasoning. A final correct-looking object does not establish that an intervention respected its constraints.

Digital brick environments offer discrete identities, hierarchical assemblies, and reproducible edits. Recent work studies generative assembly [3, 7], interactive reconstruction [9], and physical planning [5]. These settings motivate a complementary question: how can evaluation sepa-

rate the effect of object structure from the effect of the task requested on that object?

BrickAtlas crosses two axes. The structural axis ranges from D1 local components through D2 assemblies and D3 mechanisms to D4 systems. The task axis ranges from atomic questions to metacognitive/physical reasoning, executable procedures, and integrated constraints. All 48 task families occur on every configuration. This supports comparisons across levels at fixed task identity and across tasks at fixed object identity. The axes specify coverage, not an assumption that every higher-level question is empirically harder.

Three choices make failures inspectable. First, physics-derived questions refer to archived simulator observations. Second, a submitted procedure executes a public state machine, so independent legal actions can commute without matching one canonical answer string. Third, intermediate states are rendered by the original 3D application using the same interpreter as scoring. A rejected transition can thus be inspected alongside its missing precondition and unchanged geometry.

The release contains 144 configurations and 6,912 questions. Its authored, reusable constructors allow systematic coverage but create correlated examples. We distinguish configurations, geometry fingerprints, task instances, and views throughout. The public development set contains no hidden confirmation split. OMR/LDraw examples inform visual design but are not benchmark cases.

Our contributions are:

1. a complete configuration-by-task design with four structural strata and 48 families repeated on every object;
2. executable action, resource-schedule, and possible-world contracts that expose intermediate failures and information requirements;
3. a reproducible 3D release with 576 canonical views and shared scoring/replay semantics; and
4. data-derived structural statistics, answer-distribution controls, physical audit results, and a bounded historical model diagnostic.

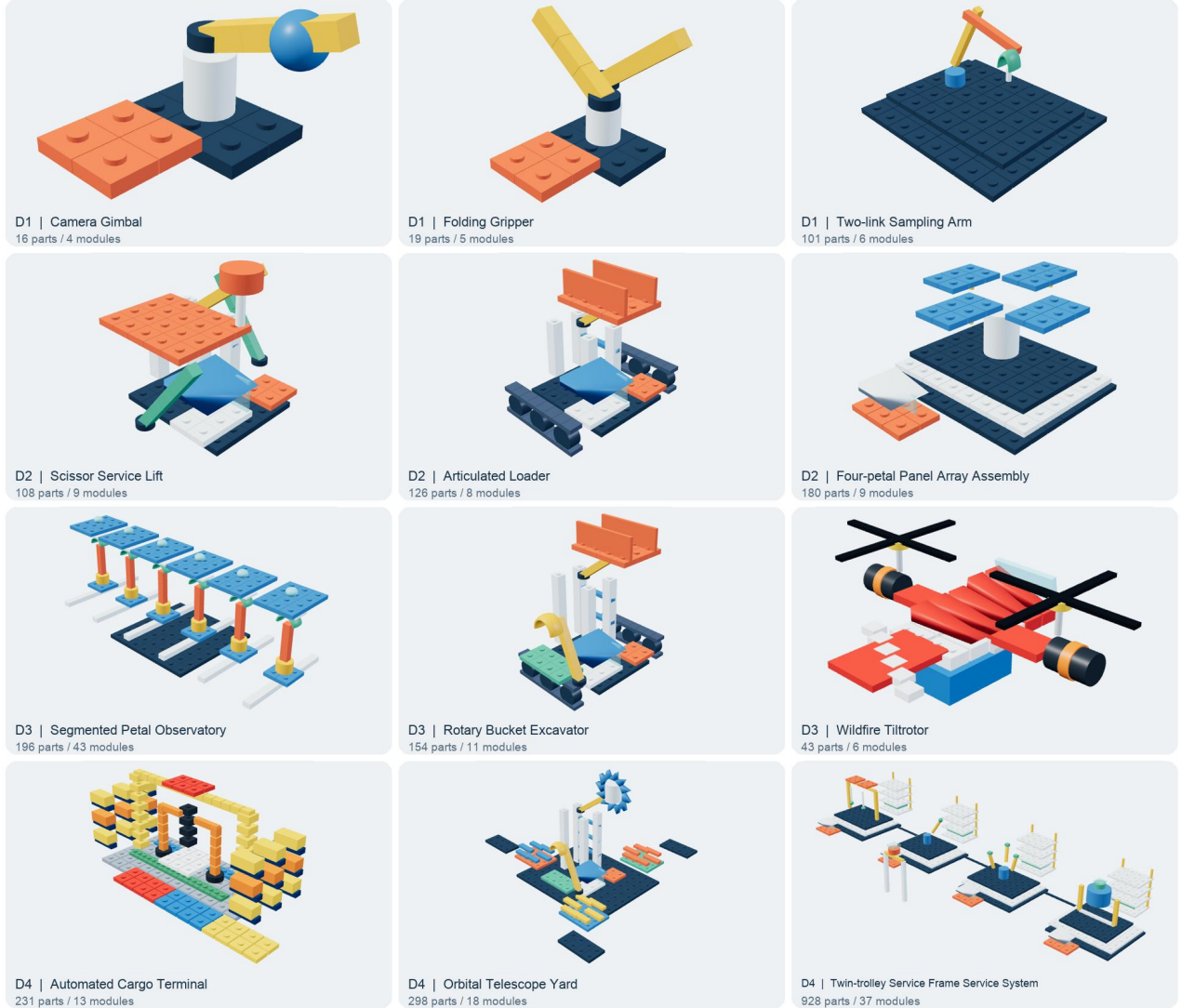


Figure 1. **Representative configurations across structural strata.** Rows progress from D1 components to D4 systems. All panels are actual 3D canvas renders. The release includes 144 configurations, and the supplement provides four views and a task index for every configuration. Part count and structural scope need not increase together.

2. Related Work

Generation and structural validity. BrickGPT [7] addresses stable brick generation. BrickNet [3] represents assembly with typed connection graphs and evaluates connectivity and collisions using valid-prefix length. It reports 320,808 pretraining samples and 67,185 supervised fine-tuning samples; these are training-sample counts, not independent source-object counts. BrECS [1] studies sequential assembly under a budget. BrickAtlas fixes the configuration and varies questions and interventions; it does not introduce a competing generator or claim comparable asset scale.

Reasoning and construction plans. LEGO-Puzzles [8] studies multistep spatial reasoning. PhyBlock [5] contains 400 assembly tasks and 2,200 VQA tasks, using Genesis and progressive planning requirements. Manual-based methods such as MEPNet [10], Neural Assembler [11], and TreeSBA [2] recover poses or construction steps. Our contribution is the shared task-by-configuration organization with executable contracts, rather than novelty of every task family.

Disassembly and embodiment. Break and Make [9] requires inspection and disassembly followed by reconstruction from an empty scene in LTRON. Its source collection

includes 1,727 OMR files; evaluation includes part, pose, and edge agreement and assembly edit distance. Our dismantling task is a dependency contract over rigid modules, and does not reproduce that visual exploration problem. WorkBenchMark [6] and BC-Bench [4] address embodied or compositional assembly. We separate digital state validity from insertion force, connector tolerances, grasping, and hardware transfer.

3. Library and Crossed Design

3.1. Representation

A model consists of parts $\mathcal{P}$, rigid modules $\mathcal{M}$, and declared joints $\mathcal{J}$. Each part records shape, dimensions, local pose, color, and module membership. Each module records world pose, mass, role, and anchoring status. Each joint records endpoints, local anchors, type, axis, and limits. Parent-to-child joint directions induce an assembly dependency order, while support queries use undirected connectivity to anchored roots.

The visible vocabulary includes bricks, plates, beams, windows, slopes, arches, cylinders, wheels, axles, gears, and spheres. These are LEGO-like digital primitives, not verified commercial part specifications. A procedure operates on rigid modules; it does not necessarily separate every visible primitive.

3.2. Structural Coverage and Provenance

D1 contains local devices such as gimbals, grippers, hatches, and valve stands. D2 introduces complete assemblies such as service lifts, loaders, trackers, and inspection craft. D3 contains articulated mechanisms including cranes, observatories, robots, and excavators. D4 combines multiple workcells in cargo terminals, docking yards, energy stations, and service hangars (Fig. 1).

Table 2 reports measured sizes. The collection contains 42,617 parts, 2,040 modules, and 1,938 joints. D2 and D3 overlap in part count because strata describe structural scope rather than psychometrically calibrated difficulty. Local questions may remain easy even in a large scene when the relevant attributes are explicit.

All configurations are agent-authored from reusable construction primitives; no independent human design study was conducted. The release retains 48 corrected configurations and their 2,304 tasks byte-for-byte, adding 96 configurations and 4,608 tasks. Twelve added constructor families contribute two configurations at each of four levels. They include arms, grippers, carousels, cross-feed stages, cradles, drums, gantries, folding panels, elastic suspensions, gates, inclined probes, and turntables.

The 96 additions have distinct geometry fingerprints after removing names, colors, and global translation. A retained latch and probe stage share geometry with different

joints, giving 143 geometry fingerprints overall. Constructor reuse still creates dependence among geometrically distinct layouts. Future train/test partitions must group shared constructors and lineage, not randomly split individual questions. The retired procedural-grid sources are excluded from current counts.

3.3. Matched Task Coverage

Table 3 defines the crossing. Task IDs bind configuration and family. Another render, color, answer order, or fault condition does not create a new source object. The 6,912 outputs comprise 4,608 single choices, 720 choice sets, 1,152 action programs, 144 schedules, and 288 policies. Procedural tasks and two integrative families share the action-program format; task layer and output format are distinct annotations.

4. Task and Scoring Contracts

A task is $t = (m, f, x, \mathcal{Y}, V)$: configuration m , family f , public evidence x , output space $\mathcal{Y}$, and deterministic validator V . An inference input contains canonical image evidence and the task-specific public record, excluding the reference answer. The browser exposes reference answers for audit and is therefore separate from the blinded input export.

4.1. Atomic and Metacognitive Questions

Atomic tasks return a choice ID. Distractors vary a relevant attribute, identity, count, set, or transformation. Option order rotates deterministically per task. Set-valued alternatives are canonicalized before deduplication; vectors and programs remain ordered. Some questions have fewer than four distinct alternatives, so chance baselines must use the actual option count.

Multiple-choice tasks require the complete feasible set. Accessibility can have an empty feasible set when every candidate path collides. Counterfactual support removes a specified module, excludes it from the answer, and recomputes reachability to any anchored root through remaining joint edges. Redundant support paths are allowed.

The base inspection family uses a uniform finite prior and deterministic observations. For query q with observation groups of size n_o among N worlds and cost $c(q)$, utility is

$$U(q) = \frac{-\sum_o (n_o/N) \log_2(n_o/N)}{c(q)}. \quad (1)$$

All maximizing queries are accepted. Abstention tests whether compatible worlds imply the same action. Posterior questions condition the prior on an observation; weighted-posterior questions supply nonuniform weights. Expected-loss questions select an action minimizing $\sum_w p(w) \ell(a, w)$. Pareto questions enumerate every non-dominated cost/mass/stiffness alternative. Force-moment

Setting	Input and objective	Primary evidence	Relation to BrickAtlas
BrickNet	Conditional generation of brick assemblies	Graph connectivity and collision prefixes	Validity of a generated structure
Break and Make	Inspect, dismantle, then reconstruct	Interactive observations and assembly agreement	Reconstruction after exploration
PhyBlock	Physical questions and assembly planning	VQA answers and simulated assembly	Progressive physical planning
BrickAtlas	Fixed configuration, 48 repeated task families	Choices, transition traces, schedules, policies	Matched reasoning and intervention contracts

Table 1. Comparison of evaluation settings, not a performance comparison. Different source units, information conditions, and validators prevent directly pooling reported dataset sizes or scores.

Level	N	Parts/layout		Modules	Depth	Tasks
		Range	Median			
D1	36	14–124	99.5	3–10	2–6	1728
D2	36	45–180	155.5	5–12	3–7	1728
D3	36	43–449	420	4–43	2–8	1728
D4	36	214–961	700.5	11–51	3–8	1728

Table 2. Measured structural coverage. Modules and dependency depth are min–max ranges. Depth counts nodes on a longest root-to-module path.

questions use explicit lever arms and forces; physical-trace questions read simulator records.

Most atomic public records disclose structural attributes needed to solve the task. Their scores therefore measure structured reasoning with auxiliary images, not isolated visual perception. Image-removal, symbolic-removal, and viewpoint ablations are required before claiming visual reasoning gains. English examples in this paper translate the task intent; the released prompts retain their original language, which must be reported in future experiments.

4.2. Executable Procedures

An action is $a = (P_a, F_a, A_a, D_a, c_a)$: required facts, forbidden facts, add effects, delete effects, and cost. Execution from state S requires $P_a \subseteq S$ and $F_a \cap S = \emptyset$, followed by

$$S' = (S \setminus D_a) \cup A_a. \quad (2)$$

The interpreter processes submitted IDs in order and stops at the first unknown action, failed precondition, forbidden state, or budget overrun. Success additionally requires all positive terminal goals and the absence of any prohibited terminal facts. A legal incomplete prefix is not a successful procedure.

Assembly places modules after their declared predecessors. Disassembly removes children before parents. Service repair requires support, opening, removal, replacement, verification, closure, and release. Compound editing substitutes a protected color operation. Guarded repair and roll-

back expose unsafe shortcuts; exclusive-tool repair removes a shared availability fact until the current intervention releases it. Independent actions may commute, so scoring checks states rather than equality with a reference list.

4.3. From Local Reasoning to Dismantling

Figure 2 illustrates how the layers relate. Initially, the atomic `remove` family asks which module is a leaf; `parent` and `degree` query the same graph. Each disassembly action requires all children of its target to be absent. Removing the foundation first fails at step one. The `prefix` family tests analogous predecessor reasoning on assembly sequences, while `counterfactual` separately asks which modules lose anchored connectivity after a specified removal. These are separate tasks on a shared object, not additional questions counted for each replay frame.

The six-node chain takes six actions to dismantle. Repairing its terminal tool instead takes seven: support, open, remove, replace, verify, close, release. Verification deletes the fault fact; release establishes the repaired goal. Support and closure are symbolic facts and need not change rendered geometry. The supplement gives every repair state and the exact precondition/effect mapping.

The same interpreter controls module visibility and color in the website. Symbolic acceptance certifies the intervention contract; it does not simulate robot motion or physical tool contact for every action. Exploded layouts are display transformations only.

4.4. Integrated Constraints

Distributed repair combines several targets under one budget and final goal. Scheduling requires nonnegative integer start times, predecessor completion, non-overlapping use of exclusive resources, and completion by a deadline. An inspection policy selects one query and maps every possible observation to an action. Validation checks all compatible worlds, not only one sampled branch, and enforces the query budget.

No LLM judge determines correctness. Empty or malformed responses remain failures. Batch scoring retains

Task layer	Families	Per level	Total	Representative requirements
Atomic	22	792	3,168	Identify, count, edit an attribute, infer topology, transform coordinates.
Meta / physical	15	540	2,160	Diagnose, interpret physical records, inspect, abstain, minimize expected loss.
Procedural	6	216	864	Assemble, dismantle, repair, compound edit, guarded repair, rollback.
Integrative	5	180	720	Repair multiple faults, share tools, schedule, choose conditional policies.
Total	48	1,728	6,912	All 48 families occur on each configuration; each level has 36 configurations.

Table 3. Crossed evaluation design. Task layers describe composition; D1–D4 describe object structure. The supplement enumerates all families.

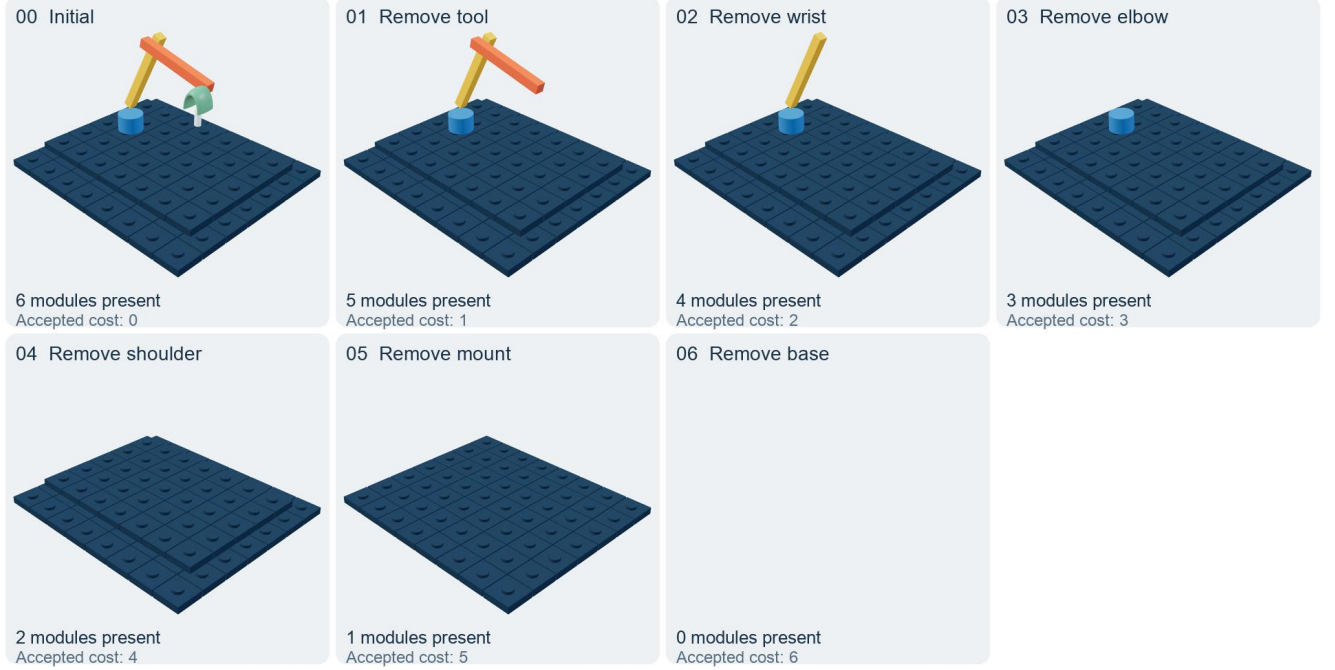


Figure 2. **Complete dependency-respecting dismantling.** Actual replay of `exp-d1-arm-2`: remove tool, wrist, elbow, shoulder, mount, then foundation. Each panel is the state after the named action, using identical crop and scale. The final panel is empty because all six modules were removed. These are symbolic removals, not continuous extraction trajectories.

missing cases in the frozen denominator and rejects unknown or duplicate case IDs. For level d and family f , the primary score is

$$A_{df} = \frac{1}{|M_d|} \sum_{m \in M_d} V(t_{mf}, \hat{y}_{mf}). \quad (3)$$

Report the 4×48 matrix before layer and overall averages. For procedures, also report accepted-prefix length, first failure cause, terminal goal satisfaction, and cost. Prefix length rewards legal incomplete programs and cannot replace exact success. A future model study should freeze cases and information conditions before inference and resample constructor families for uncertainty estimates. The present pilot is too small to support those estimates.

5. Physics and Quality Protocol

Rapier 0.20.0 runs at 120 Hz with 64 solver iterations. Stud-inclusive cuboids conservatively approximate visible parts, which form compound rigid modules. Fixed, revolute, prismatic, and spring descriptors instantiate the declared mechanism. Bases are anchored. Force-based holding motors use stiffness 60,000, damping 1,500, and maximum force 20,000 in scene units.

After 120 nominal steps, the audit measures geometric representative point displacement, angular displacement, and non-spring joint residuals. All 144 configurations satisfy thresholds of 0.35 scene units, 0.15 radians, and 0.02 scene units. Their maxima are 0.05305, 0.01825, and 0.000815 respectively (rounded upward). Every rendered vertex lies within its envelope. Initial joint anchors coincide. Oriented-cuboid queries find no inter-module pene-

Level	Choice tasks	Constant passes	Choice success (%)		Action length
			Constant	Uniform	
D1	1332	359	27.0	26.2	3–14
D2	1332	339	25.5	26.0	5–21
D3	1332	329	24.7	26.0	4–43
D4	1332	313	23.5	26.0	5–51

Table 4. Measured constant-A / empty-set control and analytic uniform choice expectation. Percentages use only choice tasks. Action length is the min–max reference-program length across action families.

tration above 0.005, including at directly joined interfaces. Joint contacts remain enabled; revolute limits are set and read back, with a driven over-limit regression control.

A separate static world performs three continuous service shape casts per object, excluding the serviced module. An off-center impulse is then followed by 120 dynamic steps and eleven sampled observations. The impulse and displacement use the same geometric representative point, preserving the response under changes of local origin. Dynamic tasks ask for the maximum of the *sampled translational response*; they do not estimate a continuous peak, rotational safety bound, or calibrated load threshold.

Finite-force servos and fixed foundations limit the meaning of nominal success. Cuboid contact validity does not establish commercial LEGO connector fit, passive stability, arbitrary motion, or insertion feasibility. Other checks cover identities, finite poses, positive dimensions, dependency order, image hashes, and website/data parity. The supplement records exact protocol settings and historical corrections.

6. Validation and Diagnostic Results

6.1. Deterministic Controls

All 6,912 public reference answers satisfy their validators. A separate solver reading public preconditions, rather than reference programs, solves all 1,152 action tasks. These are evaluator and construction checks, not learned visual-model scores; they also expose the information supplied by the symbolic input.

Table 4 separates choice-only and full-set denominators. Always selecting A for single choices and the empty set for multiple choices succeeds on 1,340/5,328 choice tasks (25.15%). Empty outputs on other contracts give 1,340/6,912 overall (19.39%). Uniform guessing has expectation $1/k$ for a single choice with k options and 2^{-k} for uniform subset selection. Varying option counts prevent a universal 25% chance baseline.

Semantic-majority frequencies measure the most frequent answer value per family after discarding option letters. They diagnose public-set answer concentration, not held-out performance. Balanced letters alone cannot rule

Level	Atomic	Meta	Procedure	Integrated
D1	1/1	0/1	1/1	0/1
D2	0/1	0/1	1/1	0/1
D3	1/1	0/1	0/1	0/1
D4	1/1	0/1	1/1	1/1

Table 5. Historical GPT-4.1 mini pilot on retained configurations. Every cell contains one fixed question; values are exact counts, not estimates of whole-cell accuracy.

out a constant semantic shortcut. The supplement reports every family’s frequency and action length where applicable. For example, the public majority is 100% for `no-op`, 90.97% for `degree`, and 83.33% for `anchor`. Thus some atomic families currently function as compliance controls and have limited discriminative value; crossing them with larger objects does not resolve their answer concentration.

6.2. Historical Model Pilot

The fixed base-48 pilot selected local-frame transformation, expected loss, guarded repair, and resource repair at each level. GPT-4.1 mini received one image and public structured evidence at temperature zero, with a 2,200-token output cap, no automatic retries, and no provider fallback. The 16 responses yield seven exact successes at a recorded cost of \$0.013621. Raw predictions, public-input hashes, provider IDs, and costs are archived. No new model calls were made for the 96 additions; these results do not measure accuracy on the full 144-configuration set.

The nine strict failures comprise four malformed action-list fields, four expected-loss mistakes, and one local-coordinate mistake. Post-hoc renaming of the four malformed fields makes their programs executable. This diagnostic separates interface compliance from planning, but the official score remains 7/16. Reporting 11/16 as a new independent experiment would conflate a post-hoc transformation with the measured system. A larger frozen study is needed to compare models, strata, or information conditions.

6.3. Inspectable Release

Every configuration opens in the original application’s 3D workspace with orbit, zoom, pan, canonical views, module isolation, explosion, and image export. Submitted or reference actions can be replayed with a step slider and first-failure feedback. Selection and isolation preserve the active modules at the selected frame.

Offline and website model records are byte-identical. The release contains 576 canonical views, four per configuration. A review dashboard supports per-task decisions, requires reasons for rejection, and exports feedback for subsequent revision. This interface provides review capability; it does not imply completed human annotation. The

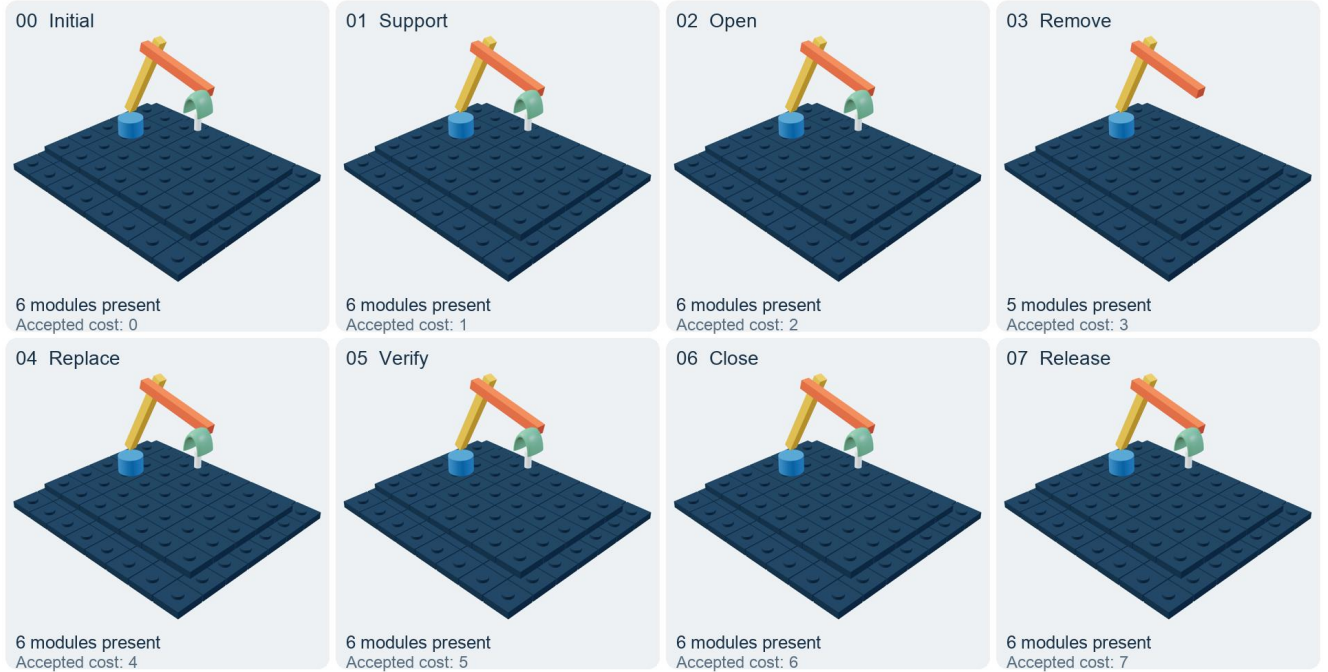


Figure 3. **Seven-step service repair on the same object.** The tool disappears after removal and returns after replacement. Support, access, verification, closure, and release update symbolic facts; visually identical frames can therefore have different legal next actions. Every frame is an actual replay state. The supplement lists the exact facts and contrasts rejected removal attempts.

supplement indexes every configuration and task family; full inputs and answers are distributed in machine-readable records.

7. Limitations

The structural strata remain authored categories. No human timing study, item-response calibration, or causal difficulty validation has been completed. Shared constructors correlate layouts; 144 configurations do not represent 144 independent mechanical families. The public development set and tiny historical pilot cannot establish robust ranking, population accuracy, or generalization.

Physical realism is limited by compound bodies, cuboid envelopes, anchored bases, and uncalibrated parameters. Declared graph edges do not certify commercial connector compatibility. Symbolic repair and exploded geometry are not collision-free robot trajectories. Structured inputs may make many tasks solvable without images. These limitations motivate family-disjoint hidden designs, manual geometry review, calibrated connectors, human measurements, and information-condition ablations.

8. Conclusion

BrickAtlas Hierarchy supports matched evaluation of brick structure, reasoning, and executable intervention. Its 144 configurations and 48 repeated families provide 6,912 in-

spectable contracts, supported by simulator records and 3D state replay. Data controls establish execution consistency; independent model and physical validation remain necessary to establish broader capability claims.

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